Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean
220423668	3.6511	4.525%	755	17.67	6"	Alias
365998904	1.2494	2.398%	375	15.52	3"	Clean
233452636	1.9063	5.467%	202	8.94	2"	Thin margin
437171483	1.4735	4.947%	923	73.37	3"	Clean
411457490	2.4324	5.531%	431	58.00	1"	Alias
295373509	5.4219	8.026%	247	22.12	2"	Clean
374158907	14.6563	4.974%	179	21.40	2"	Alias

TIC 19569810

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

```
BLS EPHEMERIS
  Period        7.006222 d
  Epoch (t0)    1495.7153 BTJD
  Duration      0.1200 d
  Depth         10.923 %  (109,228 ppm)
  SNR           156.2

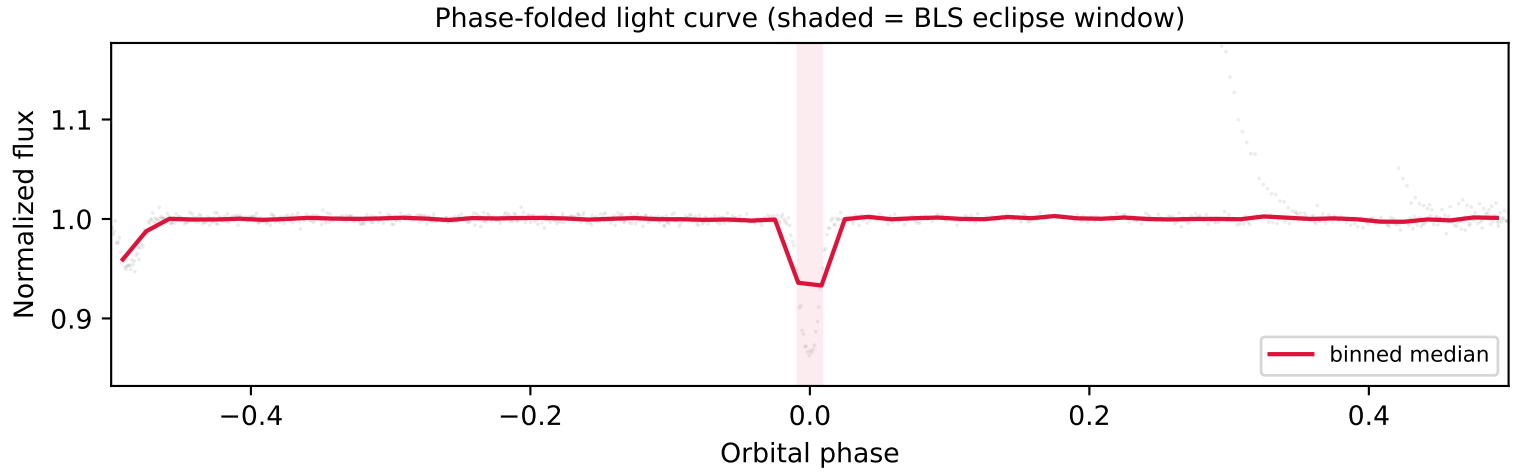
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      3  (gate >= 3)
  Robust depth ratio     1.05  (gate >= 0.50)
  Robust depth sigma     111.2  (gate >= 7.0)
  Odd/even z-score       0.46  (consistent)
  Secondary eclipse      detected, phase -0.48, 8.3 sigma
```

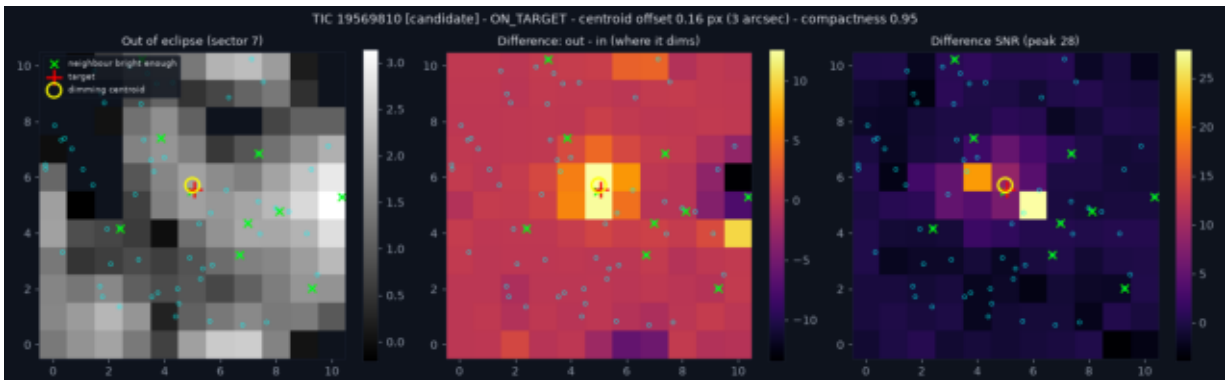
```
CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 7    light-curve dip in this sector: 115.7 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   27.94   (gate >= 5.0)      Compactness   0.95   (gate >= 0.50)
  Dimming centroid offset     0.16 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 836042204
    0.22 px away, dTmag +2.38 - farther from the dimming than the target
```





Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.052 px (dimming centroid 0.164 px from target vs 0.216 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.

TIC 349116460

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        1.296754 d
  Epoch (t0)    1492.3850 BTJD
  Duration      0.1200 d
  Depth         0.689 %  (6,886 ppm)
  SNR           109.5

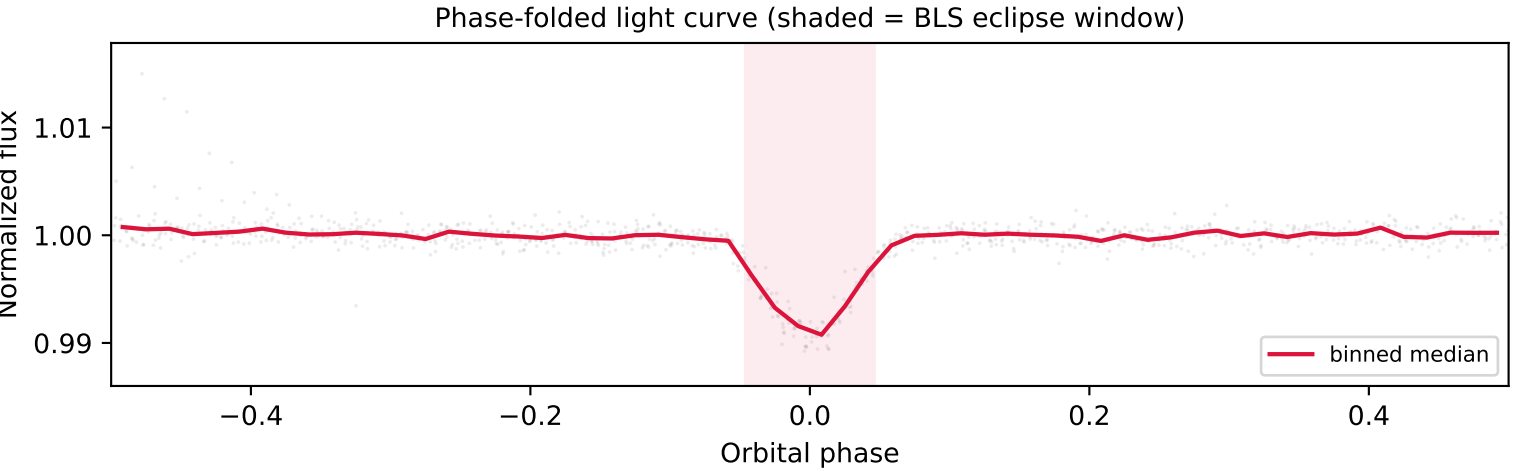
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      18  (gate >= 3)
  Robust depth ratio     0.99  (gate >= 0.50)
  Robust depth sigma     69.5  (gate >= 7.0)
  Odd/even z-score       2.66  (consistent)
  Secondary eclipse      not detected
```

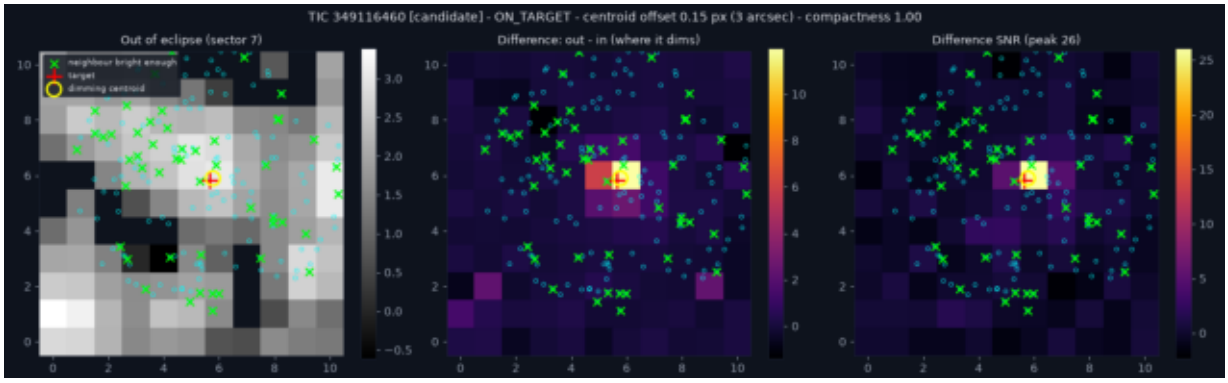
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 7    light-curve dip in this sector: 75.6 sigma    t0 refined by -0.0100 phase
  Difference-image peak SNR   26.10   (gate >= 5.0)      Compactness   1.00   (gate >= 0.50)
  Dimming centroid offset     0.15 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 349116473
    0.49 px away, dTmag +5.24 - farther from the dimming than the target
```





No caveats - clean detection on every check.

TIC 440817847

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

```
BLS EPHEMERIS
  Period       3.338136 d  (ALIASED, see below)
    -> corrected period: 6.676272 d
  Epoch (t0)   1599.3242 BTJD
  Duration     0.2440 d
  Depth        2.790 %  (27,903 ppm)
  SNR          91.8

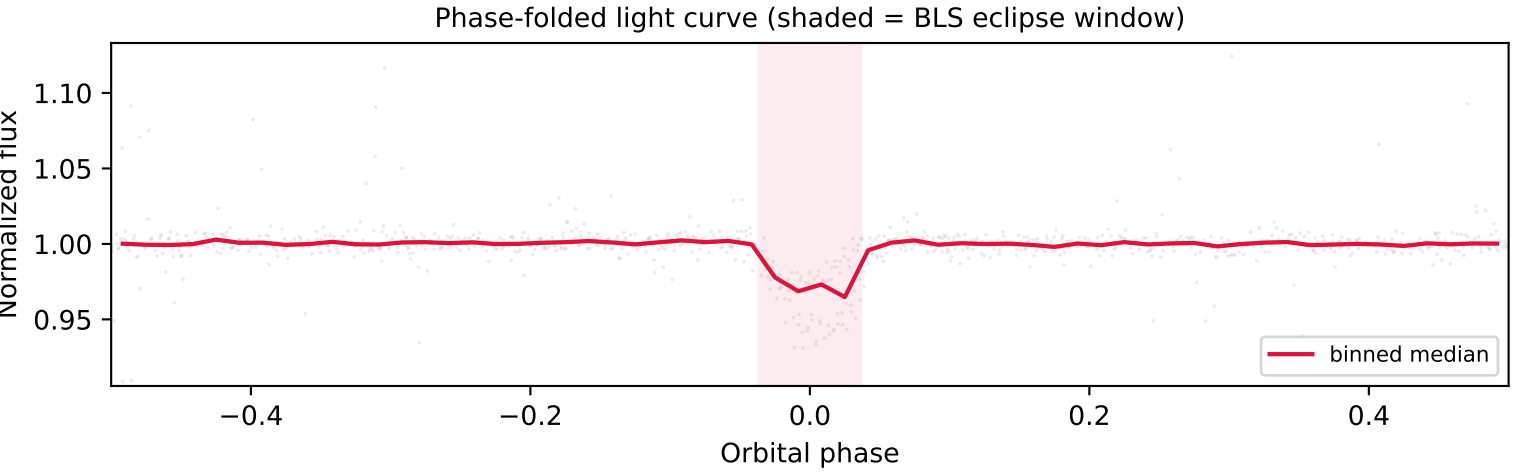
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     8  (gate >= 3)
  Robust depth ratio    0.93  (gate >= 0.50)
  Robust depth sigma    45.1  (gate >= 7.0)
  Odd/even z-score      6.44  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

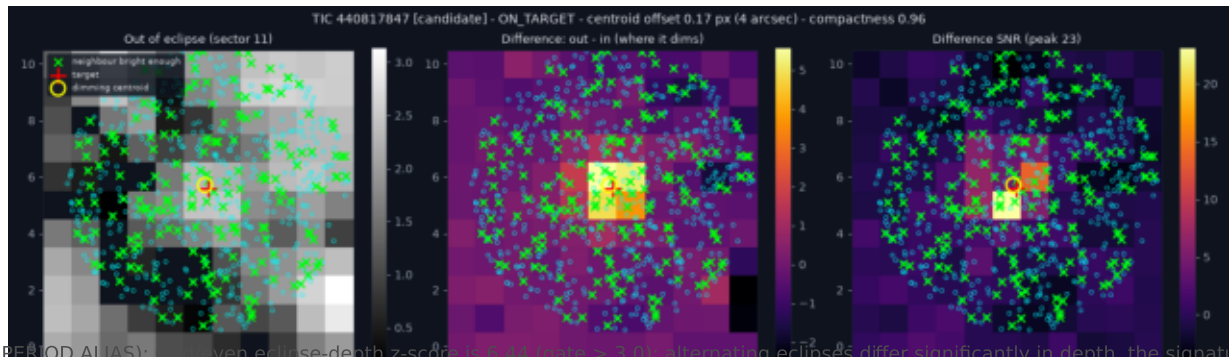
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 11    light-curve dip in this sector: 46.4 sigma    t0 refined by 0.0075 phase
  Difference-image peak SNR   23.18   (gate >= 5.0)      Compactness   0.96   (gate >= 0.50)
  Dimming centroid offset     0.17 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 440809684
    0.20 px away, dTmag +2.46 - farther from the dimming than the target
```





Caveat (PERIOD ALIAS): ... even eclipse-depth z-score is 6.44 (gate > 3.0); alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 6.676272 d, in any write-up.
Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.029 px (dimming centroid 0.168 px from target vs 0.196 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.

TIC 405529940

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        1.026596 d
  Epoch (t0)    2825.6009 BTJD
  Duration      0.0840 d
  Depth         2.998 %  (29,976 ppm)
  SNR           85.4

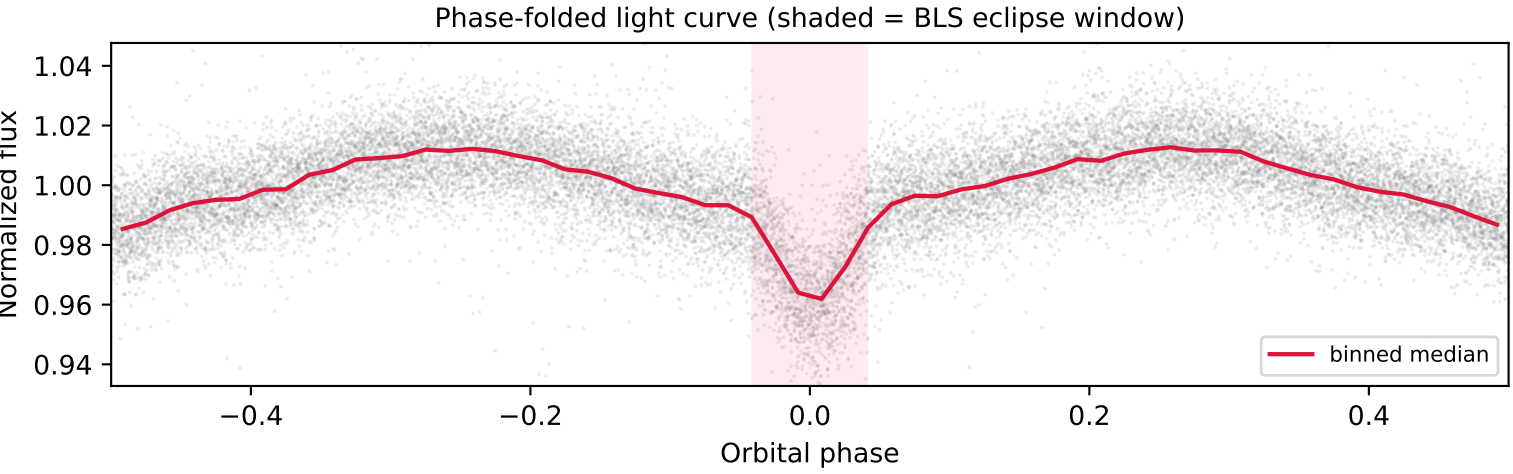
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      52  (gate >= 3)
  Robust depth ratio     1.03  (gate >= 0.50)
  Robust depth sigma     80.9  (gate >= 7.0)
  Odd/even z-score       2.42  (consistent)
  Secondary eclipse      not detected
```

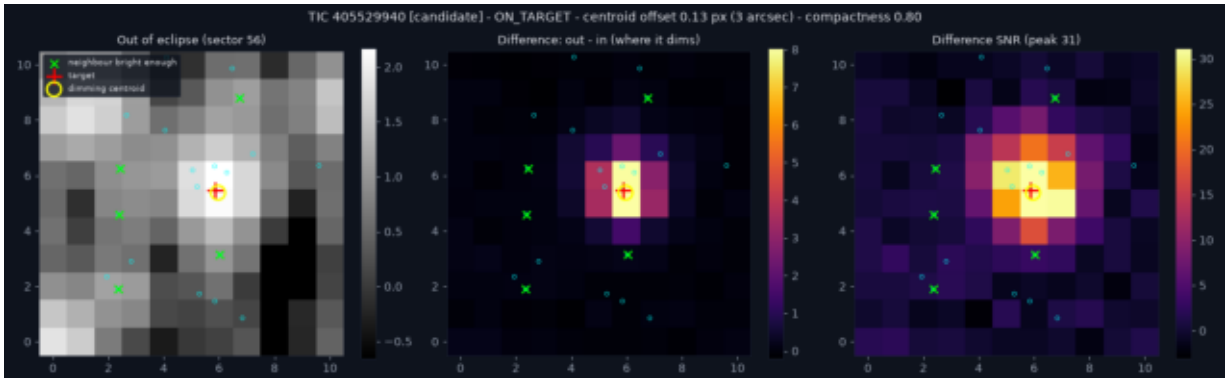
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 56    light-curve dip in this sector: 85.0 sigma    t0 refined by 0.0100 phase
  Difference-image peak SNR   31.16   (gate >= 5.0)      Compactness   0.80   (gate >= 0.50)
  Dimming centroid offset     0.13 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 118154236
    2.24 px away, dTmag +3.21 - farther from the dimming than the target
```





No caveats - clean detection on every check.

TIC 468152519

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period       5.284170 d
  Epoch (t0)   1629.3143 BTJD
  Duration     0.0580 d
  Depth        1.925 %  (19,253 ppm)
  SNR          54.4

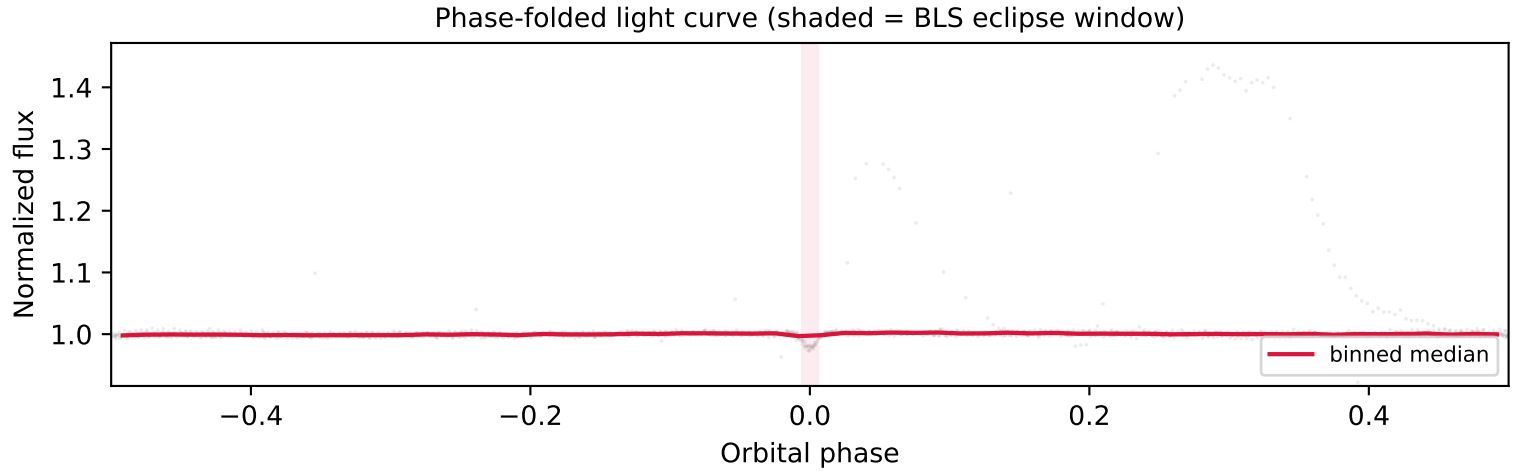
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     6  (gate >= 3)
  Robust depth ratio    1.00  (gate >= 0.50)
  Robust depth sigma    28.1  (gate >= 7.0)
  Odd/even z-score      0.01  (consistent)
  Secondary eclipse     not detected
```

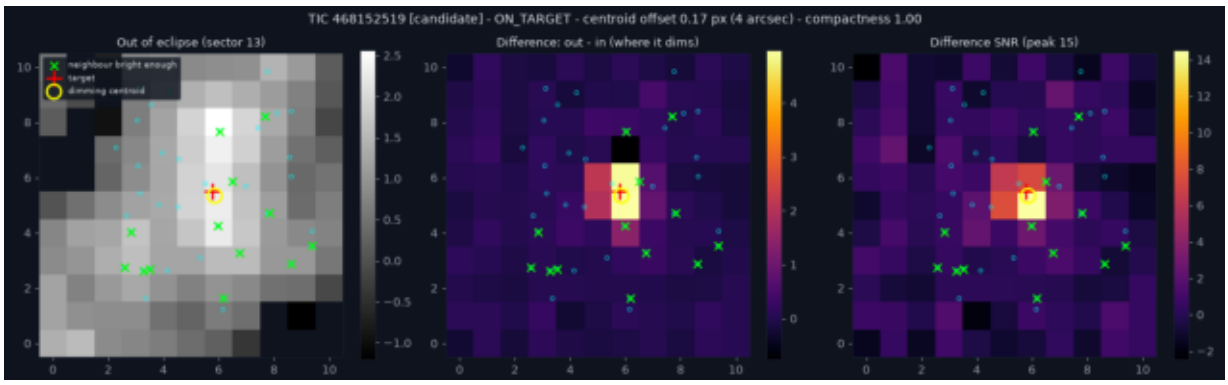
```
CATALOG CROSS-MATCH (novelty)

  VSX                                no match
  ASAS-SN                            no match
  Gaia eclipsing-binary table        no match
  Gaia variable classification       no match
  TESS EB catalog (Prsa+2022)        no match
  ExoFOP TOI list                    no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 13    light-curve dip in this sector: 18.4 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   14.51   (gate >= 5.0)      Compactness   1.00   (gate >= 0.50)
  Dimming centroid offset     0.17 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 1987652504
    0.82 px away, dTmag +4.16 - farther from the dimming than the target
```





No caveats - clean detection on every check.

TIC 257817337

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

```
BLS EPHEMERIS
  Period       5.485337 d  (ALIASED, see below)
    -> corrected period: 10.970675 d
  Epoch (t0)   3457.6769 BTJD
  Duration     0.1200 d
  Depth        0.166 %  (1,663 ppm)
  SNR          28.2

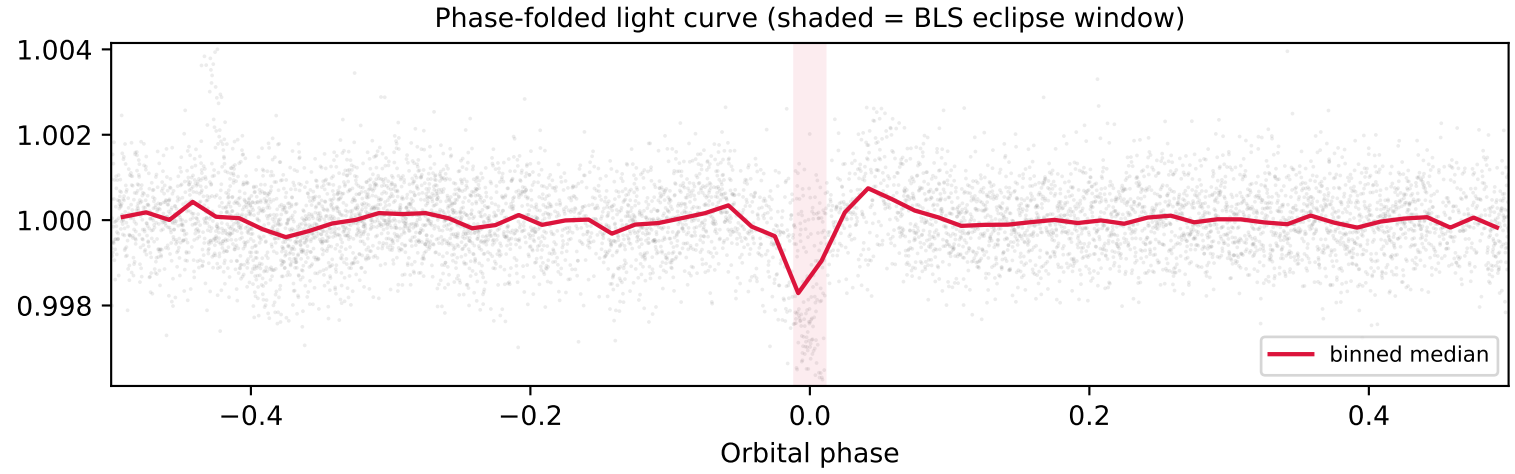
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     3  (gate >= 3)
  Robust depth ratio    1.14  (gate >= 0.50)
  Robust depth sigma    23.1  (gate >= 7.0)
  Odd/even z-score      7.82  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

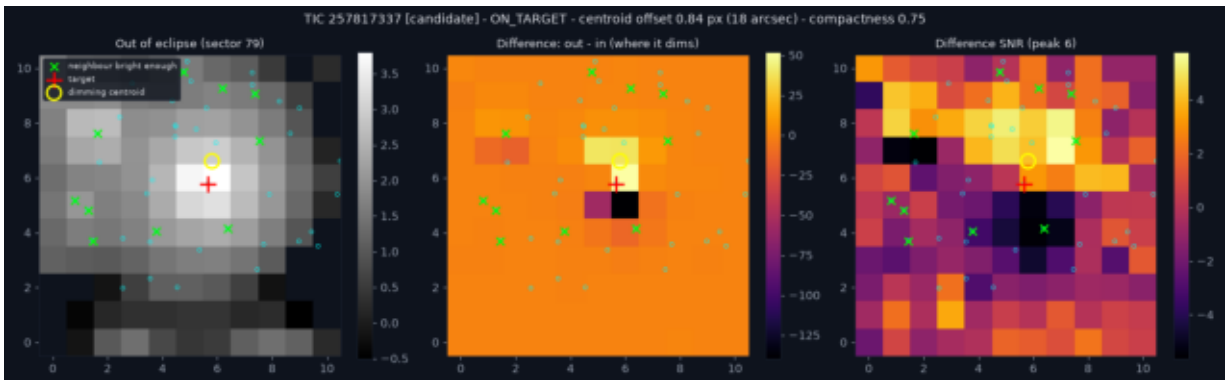
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 79    light-curve dip in this sector: 22.8 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   5.78   (gate >= 5.0)      Compactness   0.75   (gate >= 0.50)
  Dimming centroid offset     0.84 px (18 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 257817331
    1.89 px away, dTmag +6.69 - farther from the dimming than the target
```





Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 7.82 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 10.970675 d, in any write-up.